The C-Fe-Sn (Carbon-Iron-Tin) System

12.011 55.847 118.69

Introduction

Sn is known to cause temper embrittlement in steels. The increasing emphasis on recycling of scrap tends to increase the Sn content of steels. Further, in Sn plating, the structure of the surface layer determines the properties of the coating. Only limited data are available on the phase equilibria in this system. The studies of [61Sta], [82Kos], and [84Dob] covered the Fe-Fe_3C-Fe_3Sn section between 600 and 850 °C.

Binary Systems

For a description of the Fe-C phase diagram, see Section II, the C-Fe system.

The Fe-Sn phase diagram was recently reviewed by [82Kub] (Fig. 3.3 in the Co-Fe-Sn system). There are 10 invariant reactions. At 1130 °C, a monotectic reaction occurs, where the liquid, L, with 32 at.% (50 wt.%) Sn decomposes to bcc α, with 8.3 at.% (16.1 wt.%) Sn and L_2, with 69 at.% (82.5 wt.%) Sn. The liquid miscibility gap closes at about 50 at.% (68 wt.%) Sn and at ~1525 °C. At 900 °C, α and the liquid solidify through a peritectic reaction into ν_1, which is a compound of Fe and Sn, with about 43 at.% (61.6 wt.%) Sn. At 880 °C, α and ν_1 transform by a peritectoid reaction to Fe_3Sn. At 815 °C, Fe_3Sn and ν_1 form Fe_3Sn_2 by another peritectoid reaction. At 780 °C, ν_1 decomposes into Fe_3Sn_2 and the liquid. At 750 °C, Fe_3Sn decomposes eutectoidally into α and Fe_3Sn_2. At 740 °C, Fe_3Sn_2 and the liquid react to form FeSn peritectically. At 600 °C, Fe_3Sn_2 decomposes eutectoidally to α and FeSn. At 496 °C, FeSn and the liquid react peritectically to form $FeSn_2$. Just below the melting point of Sn (~232 °C), a eutectic reaction yields $FeSn_2$ and Sn. At the Fe end, the fcc γ phase closes as a loop, with the composition maximum at ~1125 °C and at 1.48 at.% Sn (3.09 wt.%). See [79Nus] for a thermodynamic analysis of this system.

Little is known about the C-Sn system. Liquid Sn dissolves a negligible amount of C.

Binary and Ternary Compounds

For crystal structure data on iron carbides, see Table II.1. In the Fe-Sn system, a number of intermediate phases are present. Fe_3Sn is isotypic with Ni_3Sn. Fe_3Sn_2 is rhombohedral. ν_1 has $B8_1$, NiAs-type structure and is stable only between 900 and 780 °C. FeSn is isotypic with CoSn. $FeSn_2$ is bct and is isotypic with Al_2Cu. Except for a small range of homogeneity of ν_1, all the compounds of this system are stoichiometric. The lattice parameters are listed in Table 11.1.

The only established ternary compound of this system is κ, with the nominal formula Fe_3SnC. [59Sta] described a number of related compounds with the general formula T_3BC, where T is a transition metal and B is either Sn or Zn. The metal atoms form an ordered fcc structure, with the C atoms occupying the octahedral voids, also in an ordered manner. Later studies preferred to denote this compound Fe_3SnC_x, with $x < 1$ [60Sta, 71Jol, 72Tre, 73Duc]. However, [82Dob] reported a homogeneity range, with $x > 1$, from $Fe_3Sn_{0.9}C_{1.4}$ to $Fe_3SnC_{1.3}$. [72Tre] reported two other ternary compounds, $Fe_3Sn_2C_y$ and $FeSnC_z$, where y and z are less than 1. However, their conclusions are based on metallographic observations, with no structural data.

The Liquidus Surface

The only available report on the liquidus surface is that of [61Sta]. Figure 11.1 shows the liquidus projection, with the primary phases of α, γ, Fe_3C(graphite), and κ (which forms through an unspecified incongruent reaction). At low Sn content, C appears as cementite; at higher Sn content, graphite forms. The data of [61Sta] are insufficient to determine the temperatures or the nature of the invariant reactions on the liquidus surface. Assuming that κ forms by the reaction L + Fe_3C(graphite) ⇄ κ and that L ⇄ α + κ originates as a pseudobinary equilibrium at a point on the monovariant curve separating κ and α, four invariant reactions—U_1, U_2, U_3, and E—are indicated in Fig. 11.1.

Invariant Equilibria

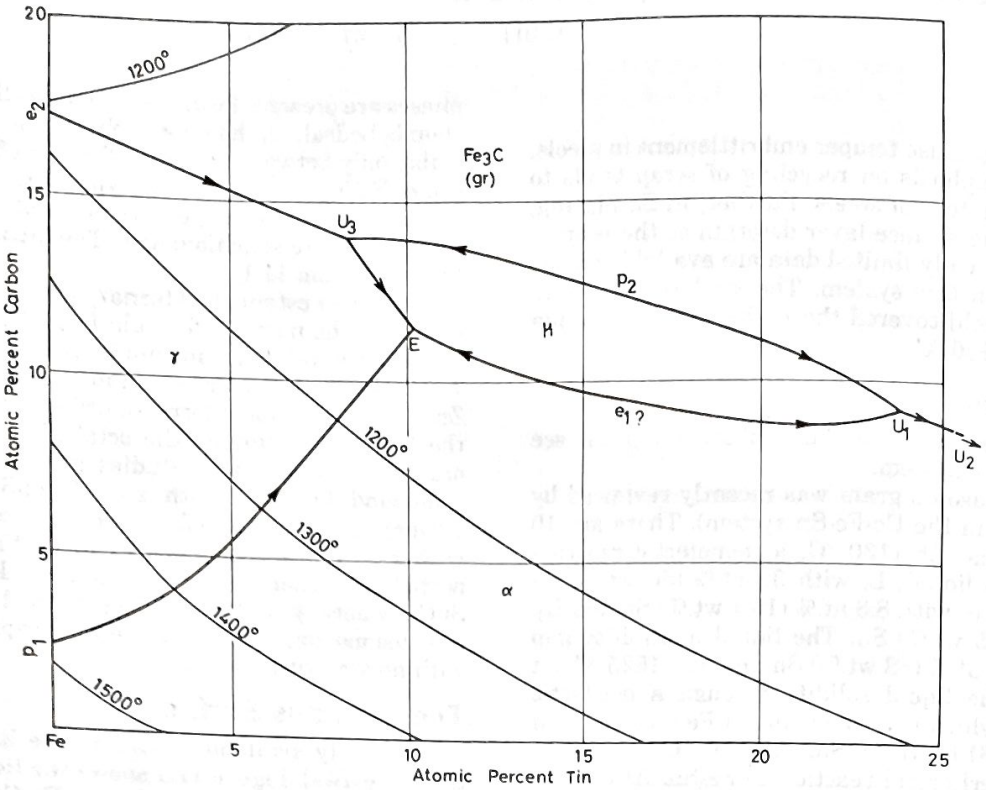
Based on the liquidus surface, a reaction scheme is shown in Fig. 11.2. The temperatures of the invariant reactions are rough estimates. The reaction scheme is extended to the solid state on the basis of speculation. The three-phase fields (γ + Fe_3C + κ) and (α + γ + κ), which

Table 11.1 C-Fe-Sn Crystal Structure and Lattice Parameter Data

Phase	Composition range	Pearson symbol	Space group	Strukturbericht designation	Prototype	Lattice parameters, nm	
						a	*c*
Fe-Sn binary; composition in at.% Sn							
Fe_3Sn	25	*hP*8	$P6_3/mmc$	$D0_{19}$	Ni_3Sn	0.5458	0.4361
Fe_3Sn_2	40	(a)	...	...	...	0.5344 (b)	1.9845 (b)
ν_1	43 to 44	*hP*4	$P6_3/mmc$	$B8_1$	NiAs	0.4230 to 0.4233	0.5208 to 0.5213
FeSn	50	*nP*6	$P6/mmm$	*B*35	CoSn	0.5300 (2)	0.4449 (2)
$FeSn_2$	66.7	*tI*12	$I4/mcm$	*C*16	Al_2Cu	0.6502	0.5315
Fe-Sn-C ternary							
Fe_3SnC_x(κ)	...	(c)	...	...	...	0.3850 to 0.3882	...

Note: For Fe-C binary data, see Table II.1.
(a) Rhombohedral. (b) Values for the hexagonal cell. (c) Ordered fcc.

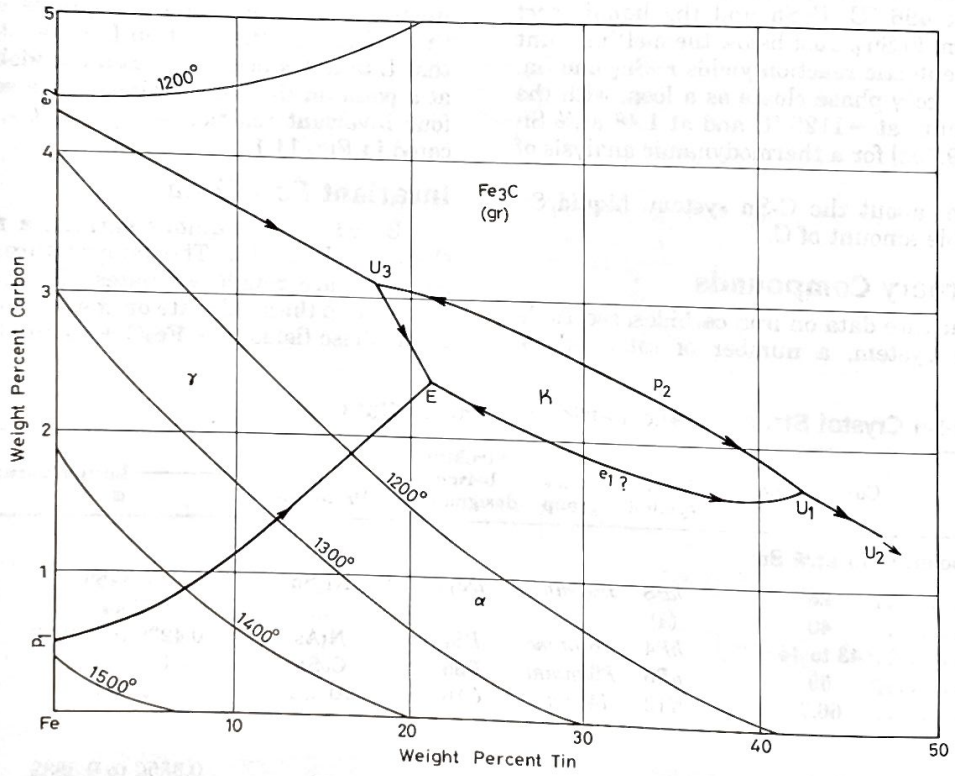
Fig. 11.1.a C-Fe-Sn Liquidus in Atomic Percent



Source [61Sta].

V. Raghavan, 1987.

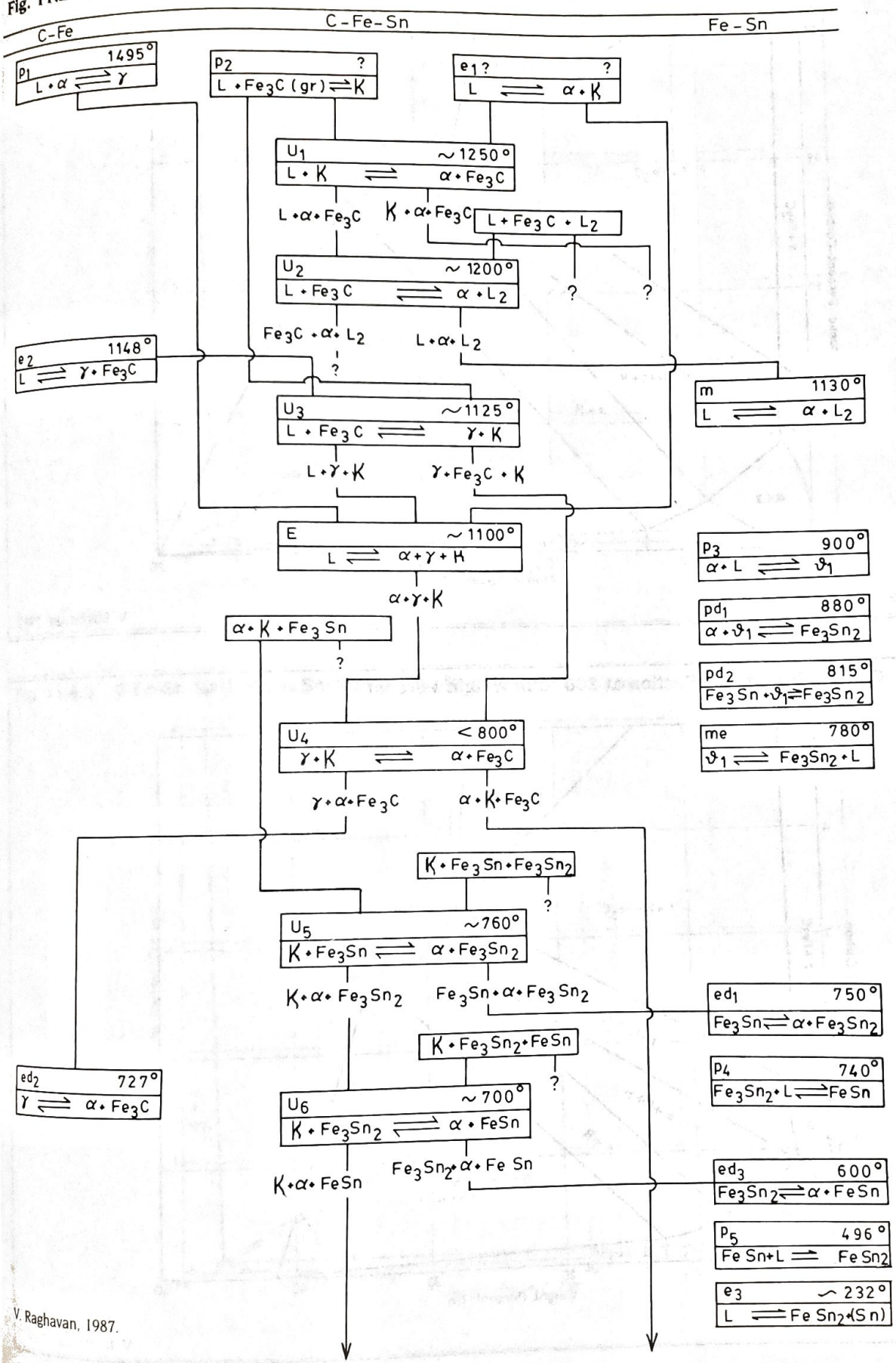
Fig. 11.1.b C-Fe-Sn Liquidus in Weight Percent



Source [61Sta].

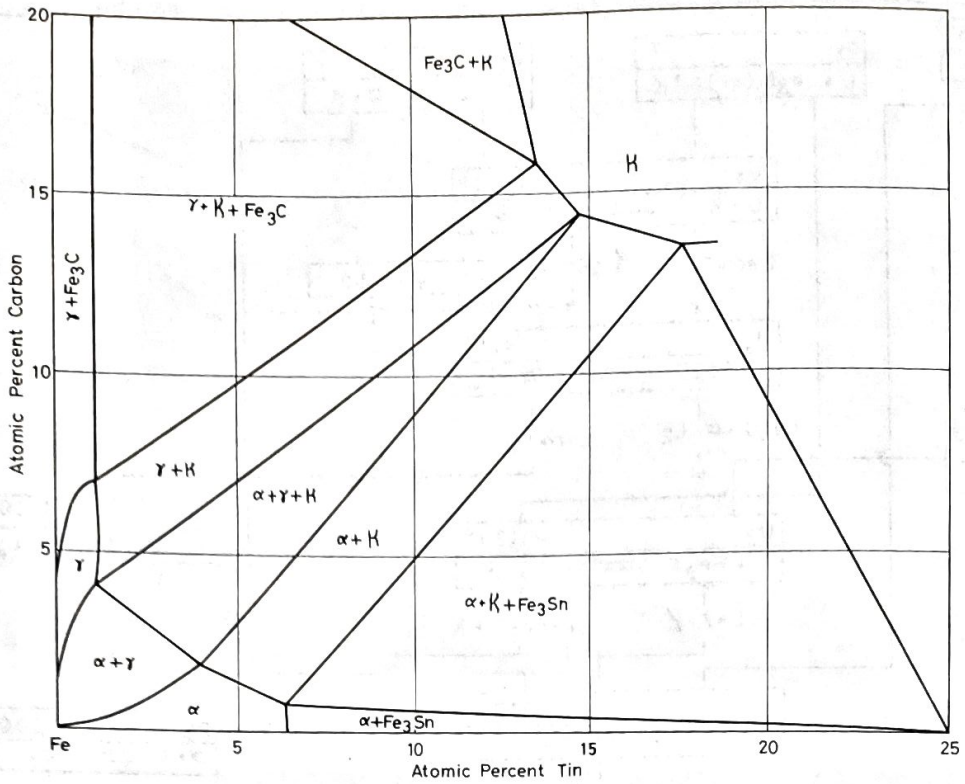
V. Raghavan, 1987.

Fig. 11.2 Reaction Scheme for Fe-Rich Alloys of the C-Fe-Sn System



V. Raghavan, 1987.

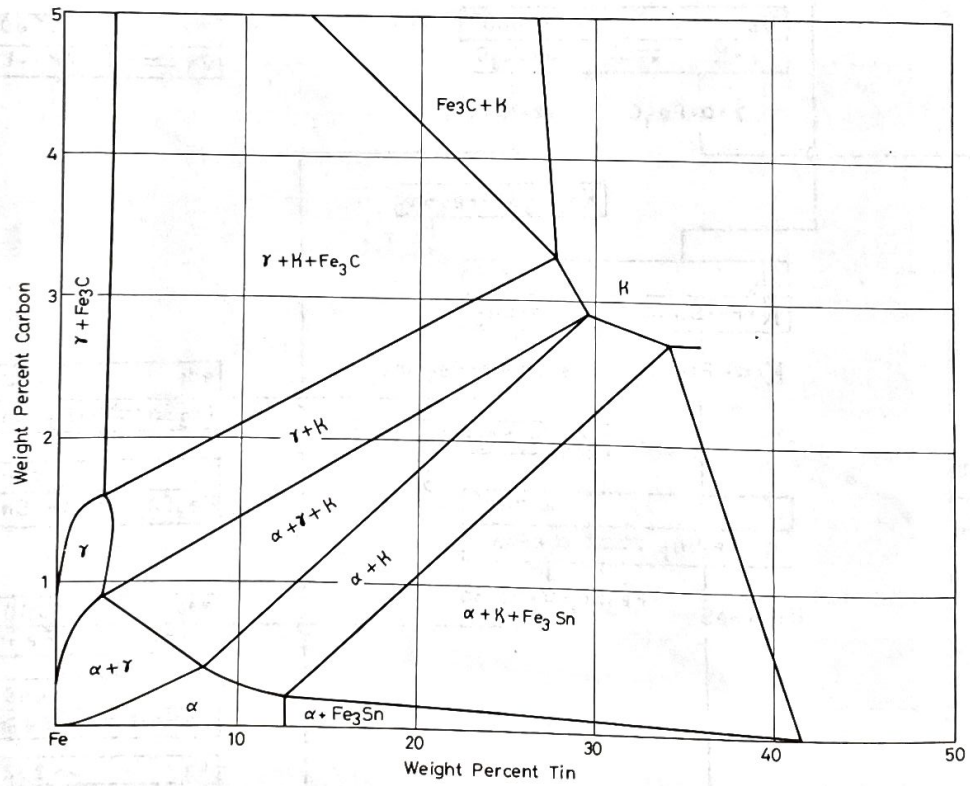
Fig. 11.3.a C-Fe-Sn Isothermal Section at 800 °C in Atomic Percent



Source [61Sta].

V. Raghavan, 1987.

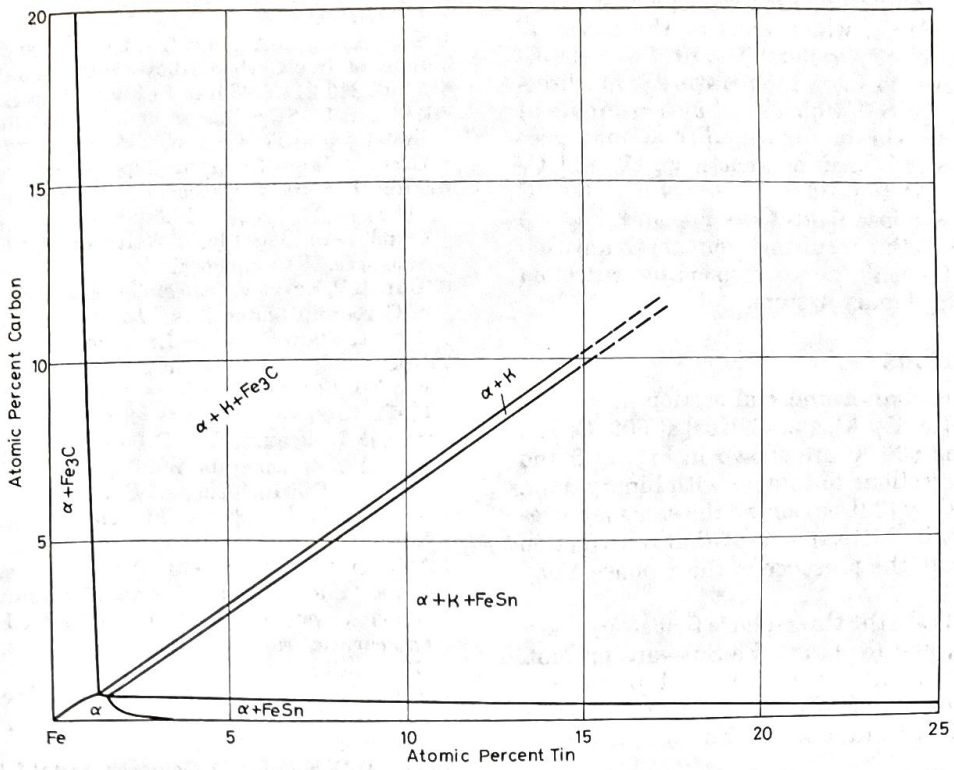
Fig. 11.3.b C-Fe-Sn Isothermal Section at 800 °C in Weight Percent



Source [61Sta].

V. Raghavan, 1987.

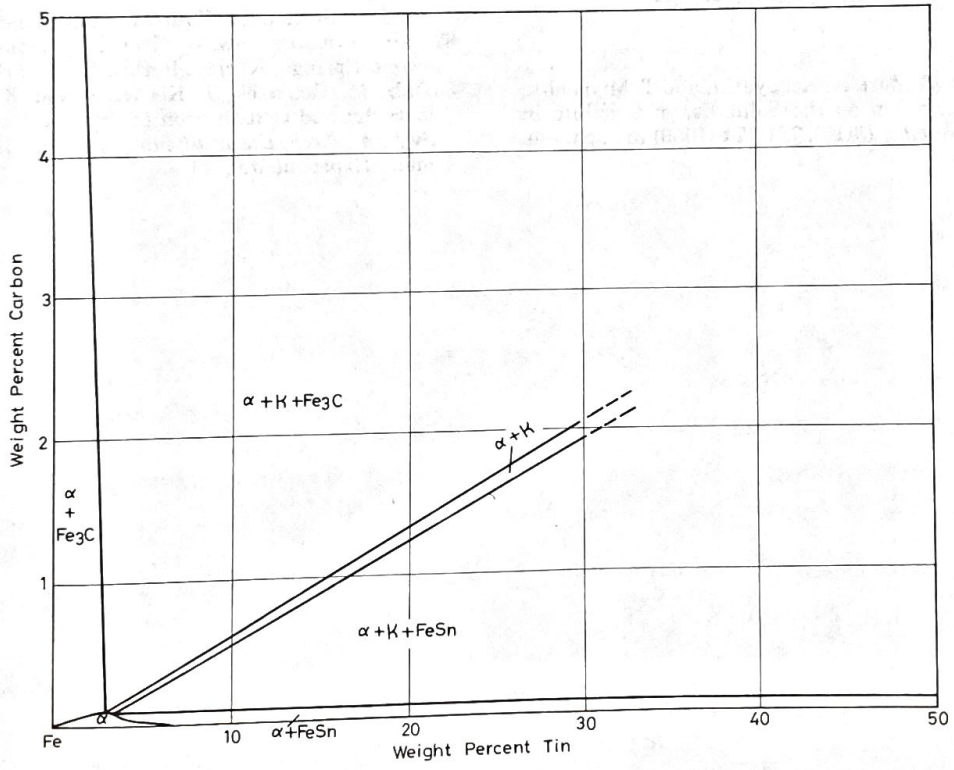
Fig. 11.4.a C-Fe-Sn Isothermal Section at 600 °C in Atomic Percent



Source [82Kos].

V. Raghavan, 1987.

Fig. 11.4.b C-Fe-Sn Isothermal Section at 600 °C in Weight Percent



Source [82Kos].

V. Raghavan, 1987.

are present at 850 °C, originate from the invariant reactions during solidification. They undergo a transition reaction, U_4, at some temperature between 800 and 727 °C to yield (α + γ + Fe_3C), which ends at the eutectoid reaction of the Fe-C binary system. The (α + κ + Fe_3C) region continues down to room temperature. The three-phase field (α + κ + Fe_3Sn), which must also originate at higher temperatures via a solid-solid reaction, goes through two successive transition reactions, U_5 and U_6 between 800 and 600 °C, to yield (α + κ + FeSn) at 600 °C and below. The three-phase fields (α + Fe_3Sn + Fe_3Sn_2) and (α + Fe_3Sn_2 + FeSn) resulting from the transition reactions U_5 and U_6 join the corresponding eutectoid reactions of the Fe-Sn binary system.

Isothermal Sections

[61Sta] constructed an isothermal section at 800 °C, [82Dob] and [84Dob] at 850 °C, and [82Kos] at 600 °C. The isotherms at 800 and 600 °C are shown in Fig. 11.3 and 11.4, with minor corrections to comply with binary data. The section at 850 °C by [82Dob] shows the same features as the one at 800 °C by [61Sta], except that a correction was needed to indicate the presence of the γ phase along the Fe-C side.

At 800 °C (Fig. 11.3), the three-phase fields—(γ + κ + Fe_3C), (α + γ + κ), and (α + κ + Fe_3Sn)—are present. Cementite is present at low Sn content and graphite at higher Sn content. At 600 °C (Fig. 11.4), the three-phase fields (α + κ + Fe_3C) and (α + κ + FeSn) are present.

Solubility of C in Fe-Sn Melts

Sn increases the activity coefficient of C in molten Fe and decreases its solubility [59Saw, 62Mor, 63Bur, 74Sig]. The first-order interaction coefficient is positive: $\epsilon_C^{Sn} = (\partial \ln \gamma_C / \partial N_{Sn}) = +20.1$ at 1550 °C in C-saturated melts. This value is applicable up to $N_{Sn} = 0.076$ [62Mor].

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